

Name-Aryan singh

sikarwar

Reg No.24BCE0403

BMAT202P – Probability and Statistics Lab

Digital Assessment 5

Two Sample Test and Chi – square Test

1. A manufacturer wants to know whether two production lines have the same average output. From Line A, a sample of 50 items has a mean output of 125 units, and the population standard deviation is 15. From Line B, a sample of 45 items has a mean output of 119 units, and the population standard deviation is 14. Test at the 1% level of significance whether Line A has a higher mean output than Line B.

Code- # Given data

```
x1 <- 125 # mean of Line A
```

```
x2 <- 119 # mean of Line B
```

```
s1 <- 15 # std dev of Line A
```

```
s2 <- 14 # std dev of Line B
```

```
n1 <- 50
```

```
n2 <- 45
```

Calculate Z statistic

```
z <- (x1 - x2) / sqrt((s1^2 / n1) + (s2^2 / n2))
```

Significance level

```
alpha <- 0.01
```

Critical value (right-tailed test)

```
z_critical <- qnorm(1 - alpha)
```

p-value

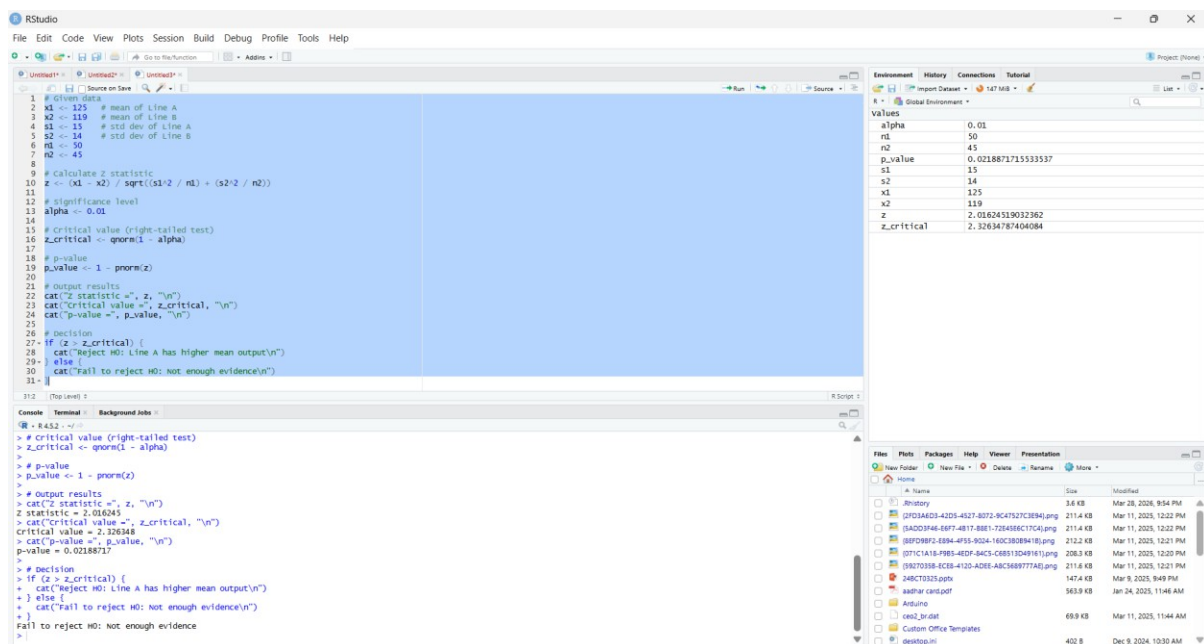
```
p_value <- 1 - pnorm(z)
```

```
# Output results
```

```
cat("Z statistic =", z, "\n") cat("Critical  
value =", z_critical, "\n") cat("p-value  
=", p_value, "\n")
```

```
# Decision
```

```
if (z > z_critical) {  
  cat("Reject H0: Line A has higher mean output\n")  
} else {  
  cat("Fail to reject H0: Not enough evidence\n")  
}
```



2. A hospital compares the recovery success rate of two treatments. In Treatment A, 86 out of 150 patients recovered successfully, while in Treatment B, 70 out of 160 patients recovered successfully. Test at the 5% level of significance whether the success proportions are different.

Code-

```
# Given data
```

```
x1 <- 86
```

```
n1 <- 150
```

```

x2 <- 70
n2 <- 160

# Sample proportions
p1 <- x1 / n1
p2 <- x2 / n2

# Pooled proportion
p <- (x1 + x2) / (n1 + n2)

# Z statistic
z <- (p1 - p2) / sqrt(p * (1 - p) * ((1/n1) + (1/n2)))

# Significance level
alpha <- 0.05

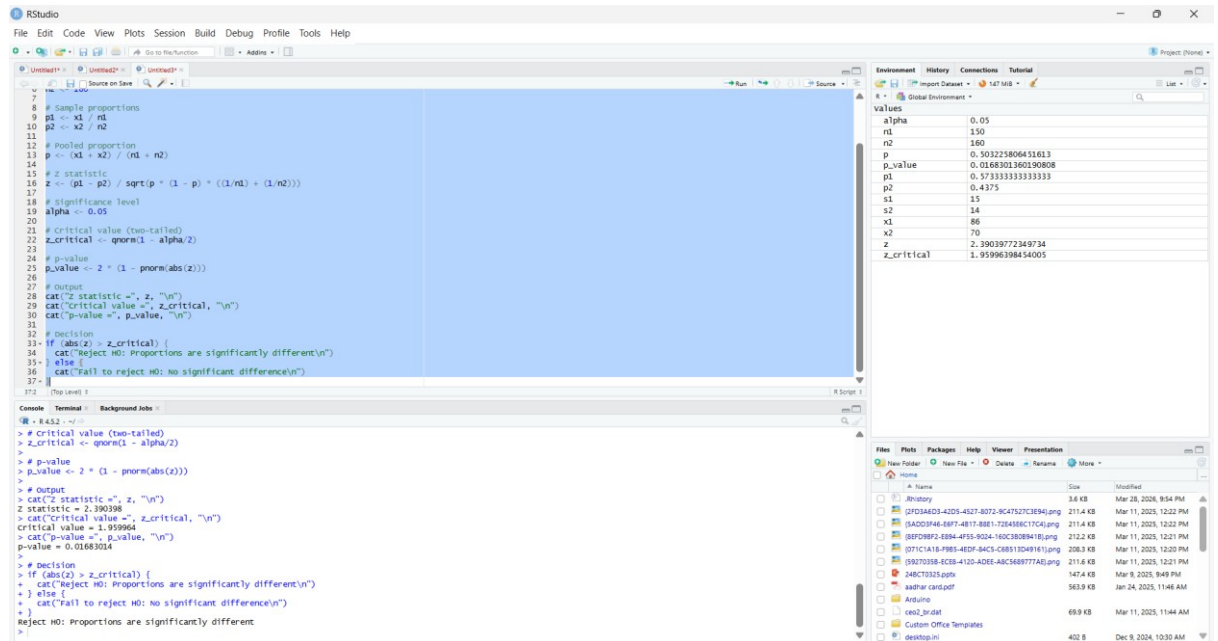
# Critical value (two-tailed)
z_critical <- qnorm(1 - alpha/2)

# p-value
p_value <- 2 * (1 - pnorm(abs(z)))

# Output
cat("Z statistic =", z, "\n") cat("Critical
value =", z_critical, "\n") cat("p-value
=", p_value, "\n")

# Decision
if (abs(z) > z_critical) {
  cat("Reject H0: Proportions are significantly different\n")
} else {
  cat("Fail to reject H0: No significant difference\n")
}

```



3. A researcher compares reaction times from two training programs. Program A is applied to 8 participants, and the times are 2.8, 3.1, 2.9, 3.0, 3.2, 2.7, 3.0, and 2.8 seconds. Program B is applied to another 8 participants, and the times are 3.4, 3.7, 3.5, 3.8, 3.6, 3.3, 3.5, and 3.4 seconds. Test at the 5% level of significance whether Program A gives a lower mean reaction time.

Code-

Data

```
A <- c(2.8, 3.1, 2.9, 3.0, 3.2, 2.7, 3.0, 2.8)
```

```
B <- c(3.4, 3.7, 3.5, 3.8, 3.6, 3.3, 3.5, 3.4)
```

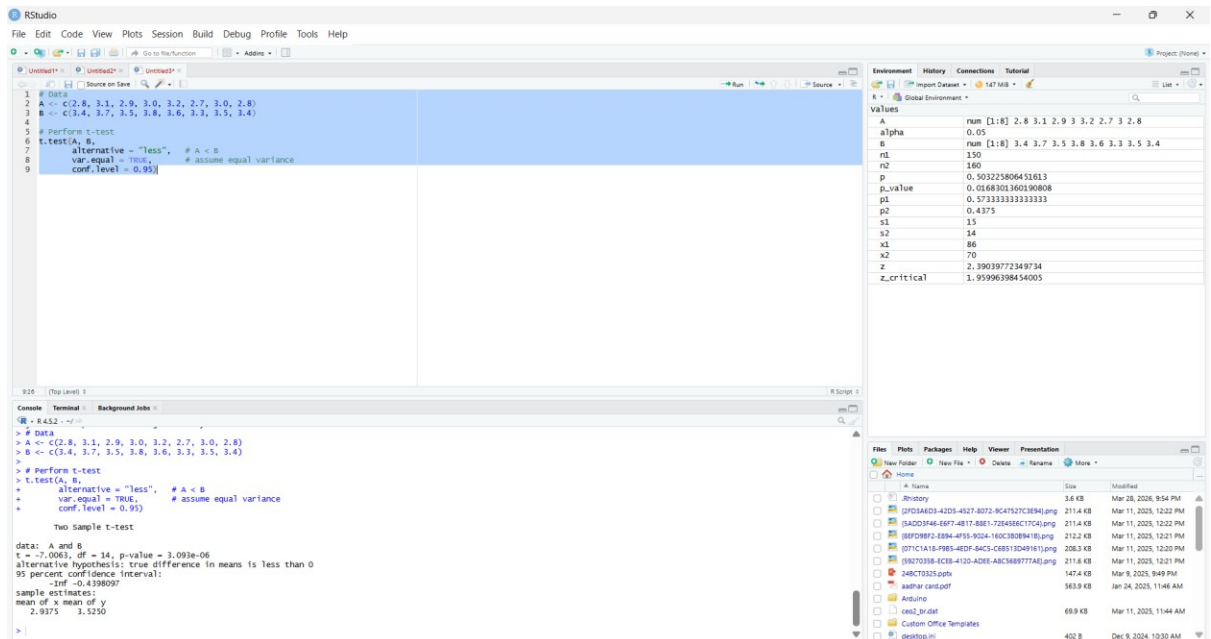
Perform t-test

```
t.test(A, B,
```

```
  alternative = "less", # A < B
```

```
  var.equal = TRUE,    # assume equal variance
```

```
  conf.level = 0.95)
```



4. A public health researcher wants to know whether mobile phone usage habit is related to sleep quality among college students. A sample of students is classified into three phone-usage groups: low, moderate, and high. Their sleep quality is recorded as good or poor, and the following table is observed: low usage has 28 good and 12 poor sleepers, moderate usage has 24 good and 16 poor sleepers, and high usage has 18 good and 22 poor sleepers. Test whether phone usage habit and sleep quality are independent at the 5% level.

Code-

Create contingency table data

```

<- matrix(c(28,12,
            24,16,
            18,22),
          nrow = 3, byrow
          = TRUE)

```

Add row/column names (optional) rownames(data)

```

<- c("Low", "Moderate", "High")

```

```

colnames(data) <- c("Good", "Poor")

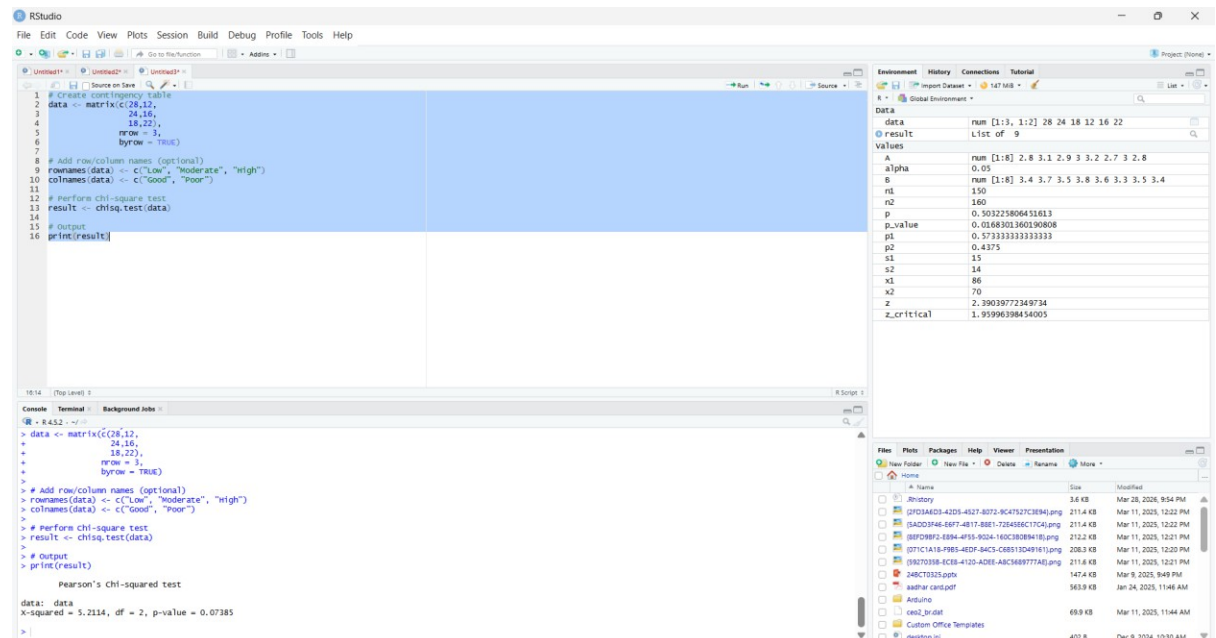
```

Perform Chi-square test

```
result <- chisq.test(data)
```

```
# Output
```

```
print(result)
```



5. A food brand claims that its snack packs are filled in the proportions 20%, 20%, 20%, 20%, and 20% across five flavor categories. In a quality-control check, 100 packets are inspected and the observed counts are 18, 22, 19, 21, and 20. Test whether the observed distribution agrees with the claimed distribution at the 5% level.

Code-

```
# Observed frequencies
```

```
observed <- c(18, 22, 19, 21, 20)
```

```
# Expected proportions
```

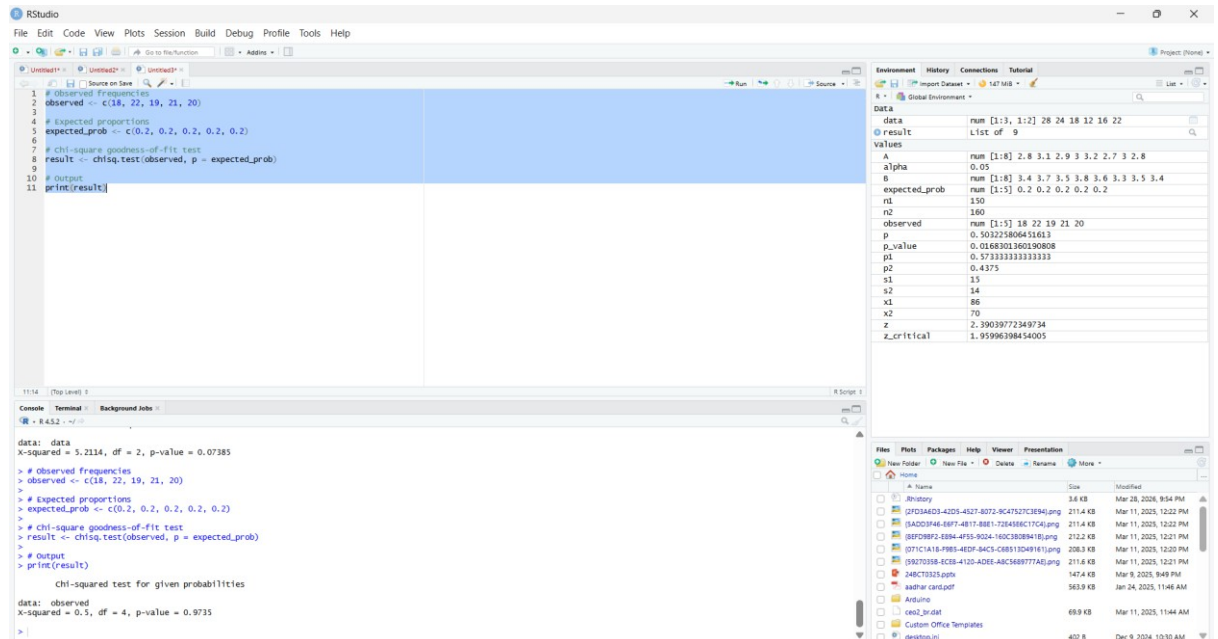
```
expected_prob <- c(0.2, 0.2, 0.2, 0.2, 0.2)
```

```
# Chi-square goodness-of-fit test
```

```
result <- chisq.test(observed, p = expected_prob)
```

```
# Output
```

```
print(result)
```



6. A biotechnology lab studies the number of successful gene edits in 10 trials for each sample. The lab believes the number of successes follows a binomial distribution with $n = 10$ and success probability $p = 0.4$. A sample of 200 experiments gives the following observed frequencies for 0 to 10 successes: 5, 14, 23, 31, 36, 34, 25, 16, 9, 5, and 2. Test whether a binomial distribution with $n = 10$, $p = 0.4$ fits the data well.

Code-

Observed frequencies

```
observed <- c(5,14,23,31,36,34,25,16,9,5,2)
```

Parameters n

```
<- 10
```

```
p <- 0.4
```

```
N <- 200
```

Possible values (0 to 10)

```
x <- 0:10
```

Expected probabilities using binomial prob

```
<- dbinom(x, size = n, prob = p)
```

Expected frequencies

```
expected <- N * prob
```

Combine small expected frequencies if needed

(Chi-square requires expected ≥ 5 ideally)

```
observed_adj <- c(observed[1:8], sum(observed[9:11]))
expected_adj <- c(expected[1:8], sum(expected[9:11]))
```

```

# Chi-square test
chisq <- sum((observed_adj - expected_adj)^2 / expected_adj)

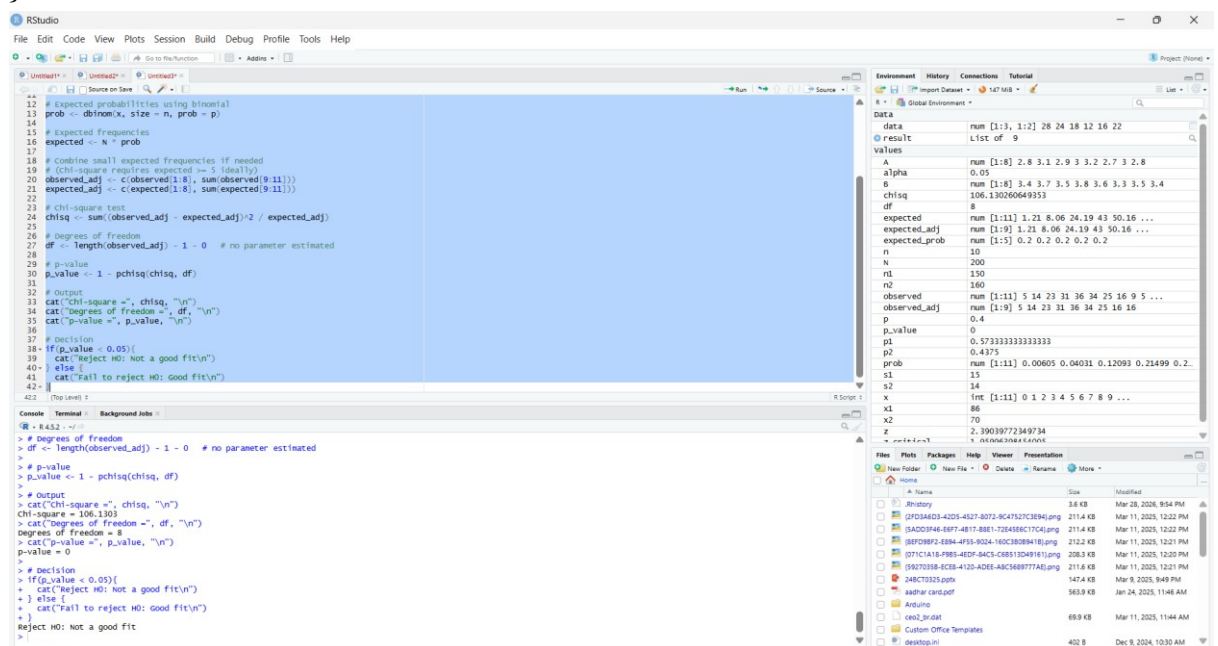
# Degrees of freedom
df <- length(observed_adj) - 1 - 0 # no parameter estimated

# p-value
p_value <- 1 - pchisq(chisq, df)

# Output
cat("Chi-square =", chisq, "\n")
cat("Degrees of freedom =", df, "\n")
cat("p-value =", p_value, "\n")

# Decision
if(p_value < 0.05){
  cat("Reject H0: Not a good fit\n")
} else {
  cat("Fail to reject H0: Good fit\n")
}

```



7. A call center records the number of customer complaints arriving per hour. The manager believes the number of complaints follows a Poisson distribution with mean $\lambda = 2$. Over 150 hours, the observed frequencies of 0, 1, 2, 3, 4, and 5 or more complaints are 18, 34, 41, 30, 17, and 10. Test whether the Poisson model fits the data at the 5% level.

Code-

Observed frequencies

```

observed <- c(18, 34, 41, 30, 17, 10)

# Parameters lambda
<- 2
N <- 150

# Values (0 to 4)
x <- 0:4

# Poisson probabilities prob
<- dpois(x, lambda)

# Expected frequencies for 0–4
expected <- N * prob

# Expected for 5 or more
expected_5plus <- N * (1 - sum(prob))

# Combine expected
expected <- c(expected, expected_5plus)

# Chi-square statistic
chisq <- sum((observed - expected)^2 / expected)

# Degrees of freedom
# df = categories - 1 - parameters estimated df
<- length(observed) - 1 - 0 # lambda given

# p-value
p_value <- 1 - pchisq(chisq, df)

# Output
cat("Chi-square =", chisq, "\n")
cat("Degrees of freedom =", df, "\n")
cat("p-value =", p_value, "\n")

# Decision
if(p_value < 0.05){
  cat("Reject H0: Poisson is not a good fit\n")
} else {
  cat("Fail to reject H0: Poisson fits well\n")
}

```

RStudio

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Source

```
13 # Expected frequencies for 0-4
14 expected <- N * prob
15 # Expected for 5 or more
16 expected_splus <- N * (1 - sum(prob))
17 # combine expected
18 expected <- c(expected, expected_splus)
19 # Chi-square statistic
20 chisq <- sum((observed - expected)^2 / expected)
21 # degrees of freedom
22 df <- categories - 1 - parameters_estimated
23 df <- length(observed) - 1 - 0 # lambda given
24 # p-value
25 p_value <- 1 - pchisq(chisq, df)
26 # Output
27 cat("Chi-square =", chisq, "\n")
28 cat("degrees of freedom =", df, "\n")
29 cat("p-value =", p_value, "\n")
30 # Decision
31 if(p_value < 0.05){
32   cat("Reject H0: Poisson is not a good fit\n")
33 } else {
34   cat("Fail to reject H0: Poisson fits well\n")
35 }
36
```

Console

```
R 4.5.2 >
> # df = categories - 1 - parameters estimated
> df <- length(observed) - 1 - 0 # lambda given
>
> # p-value
> p_value <- 1 - pchisq(chisq, df)
>
> # Output
> cat("Chi-square =", chisq, "\n")
Chi-square = 3.102836
> cat("degrees of freedom =", df, "\n")
degrees of freedom = 5
> cat("p-value =", p_value, "\n")
p-value = 0.6841352
>
> # Decision
> if(p_value < 0.05){
+   cat("Reject H0: Poisson is not a good fit\n")
+ } else {
+   cat("Fail to reject H0: Poisson fits well\n")
+ }
Fail to reject H0: Poisson fits well
>
```

Environment

Global Environment

Data

data	run [1:3, 1:2]	28	24	18	12	16	22
alpha	run [1:8]	2.8	3.1	2.9	3.2	2.7	3.2
0.05							
B	run [1:8]	3.4	3.7	3.5	3.8	3.3	3.5
chisq							
df							
expected	run [1:6]	20.3	40.6	40.6	27.1	13.5	...
expected_splus							
expected_adj	run [1:9]	1.21	8.06	24.19	43	50.16	...
expected_prob	run [1:5]	0.2	0.2	0.2	0.2	0.2	
lambda							
n							
N							
nt							
n2							
observed	run [1:6]	18	34	41	30	17	10
observed_adj	run [1:9]	5	14	23	11	36	25
p							
p_value							
p1							
p2							
prob	run [1:5]	0.1353	0.2707	0.2707	0.1804	0.0902	
s1							
s2							
x	run [1:5]	0	1	2	3	4	
x1							
x2							

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Name	Size	Modified
.Rhistory	3.6 KB	Mar 28, 2025, 9:54 PM
(2FD346D5-43D5-4527-8072-8C47527C3E94).png	211.4 KB	Mar 11, 2025, 12:22 PM
(5A0D3F46-8877-4977-8881-7284566C17C4).png	211.4 KB	Mar 11, 2025, 12:22 PM
(8B7D38F2-8294-4955-8024-160308941610).png	212.2 KB	Mar 11, 2025, 12:21 PM
(D71C1A18-F985-4EDF-8AC5-C68513D49161).png	208.3 KB	Mar 11, 2025, 12:20 PM
(59270338-EC38-412D-ADEE-ABC5689777AE).png	211.6 KB	Mar 11, 2025, 12:21 PM
248C70325.png	147.4 KB	Mar 9, 2025, 9:49 PM
aadhar card.pdf	563.9 KB	Jan 24, 2025, 11:46 AM
Audio		
cert2.dat	68.9 KB	Mar 11, 2025, 11:44 AM
Custom Office Templates		
desktop.ini	402 B	Dec 9, 2024, 10:30 AM